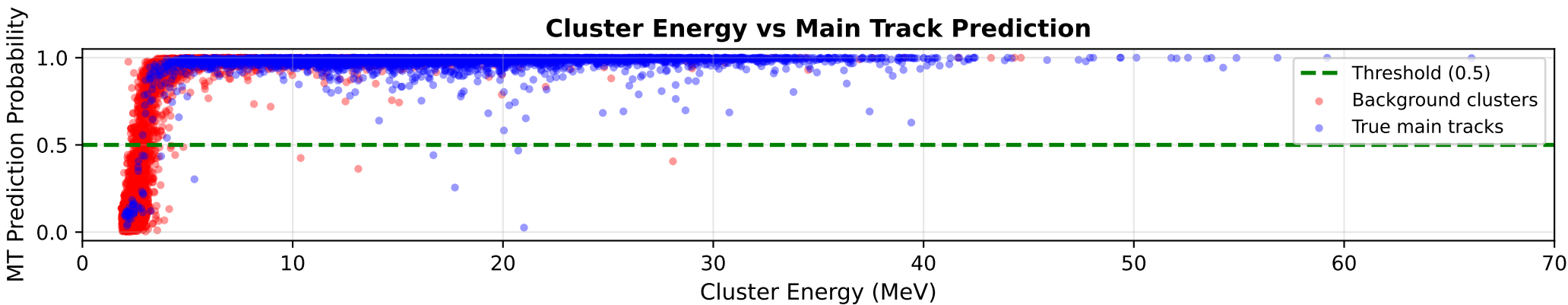


MT Prediction Analysis - Energy vs Prediction



MT INFERENCE SUMMARY - CAT000001

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DATASET:

Total clusters:	19,384
True main tracks:	3,789
True backgrounds:	15,595

PREDICTIONS:

Predicted main tracks:	6,814
Predicted backgrounds:	12,570

CONFUSION MATRIX:

True Positives (TP):	3,742
False Positives (FP):	3,072
False Negatives (FN):	47
True Negatives (TN):	12,523

PERFORMANCE METRICS:

Accuracy:	83.9%
Precision:	54.9%
Recall (Efficiency):	98.8%
Contamination (FP rate):	45.1%

ENERGY ANALYSIS:

True Positives:

Mean:	17.93 MeV
Median:	16.91 MeV
Range:	2.87 - 66.06 MeV

False Positives:

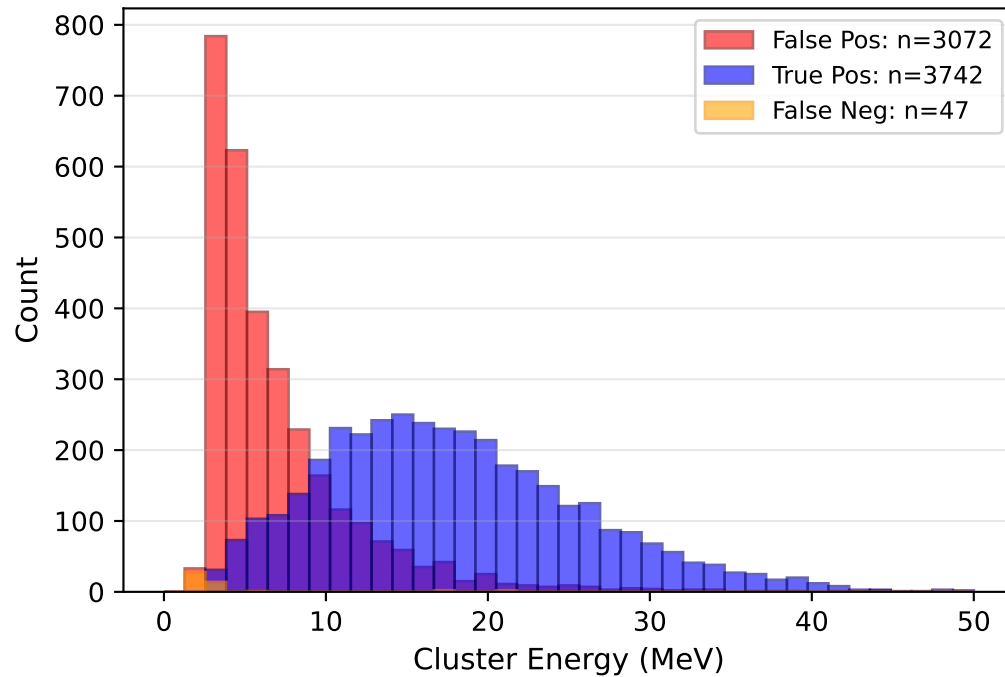
Mean:	7.03 MeV
Median:	5.44 MeV
Range:	2.18 - 44.63 MeV

LOW ENERGY ANALYSIS (< 5 MeV):

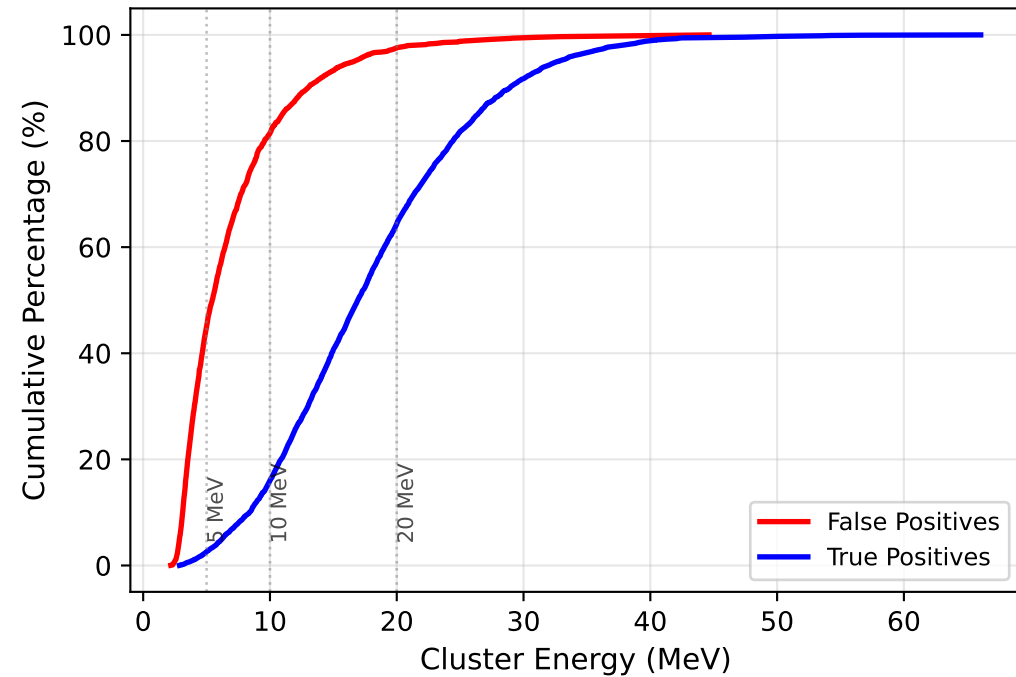
FP below 5 MeV:	1,385 (45.1% of all FP)
TP below 5 MeV:	98 (2.6% of all TP)

MT Prediction Analysis - Detailed Distributions

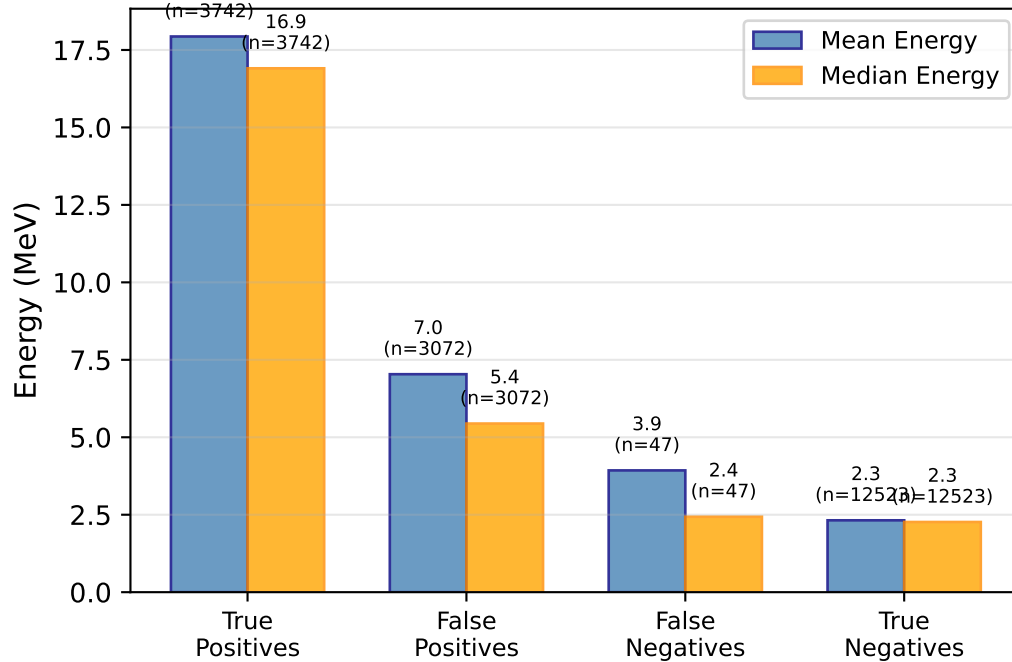
Energy Distribution by Prediction Outcome



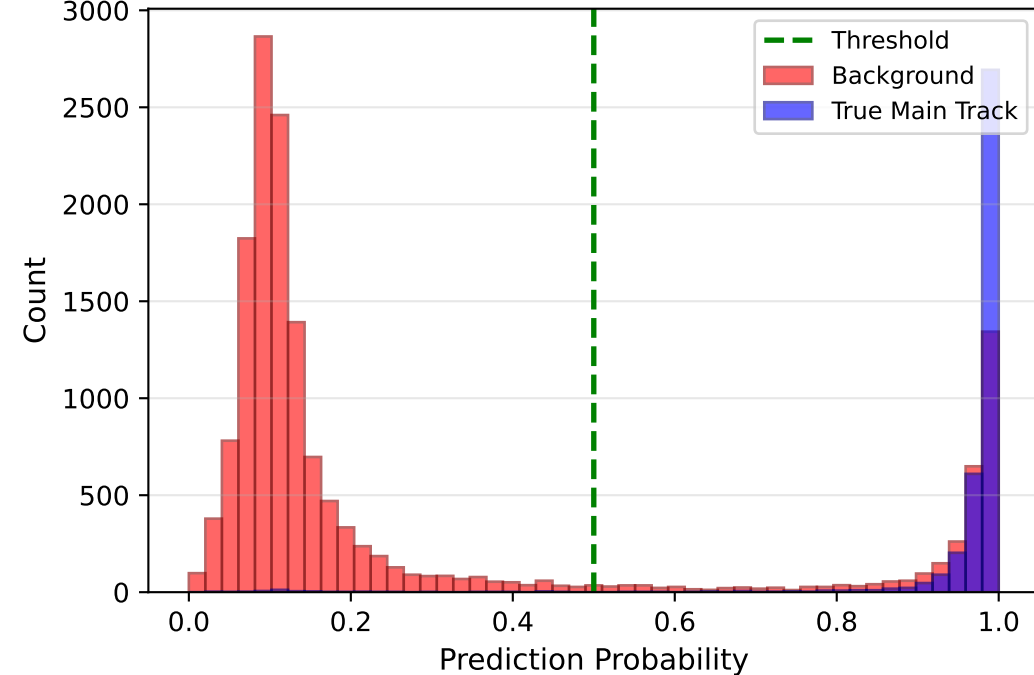
Cumulative Energy Distribution



Average Cluster Energy by Category

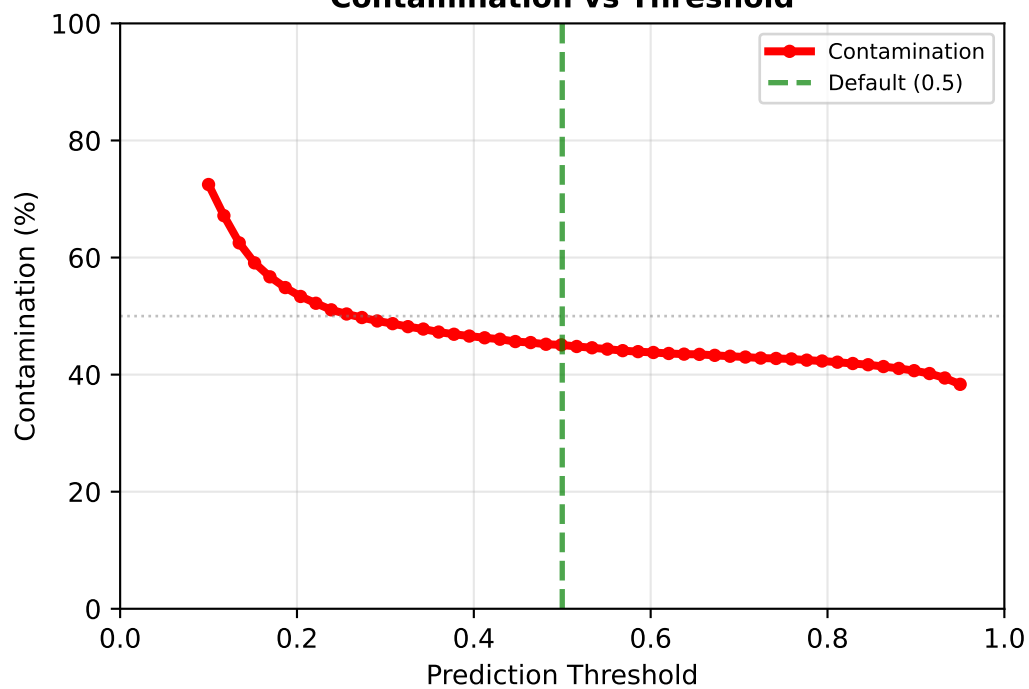


Distribution of MT Predictions

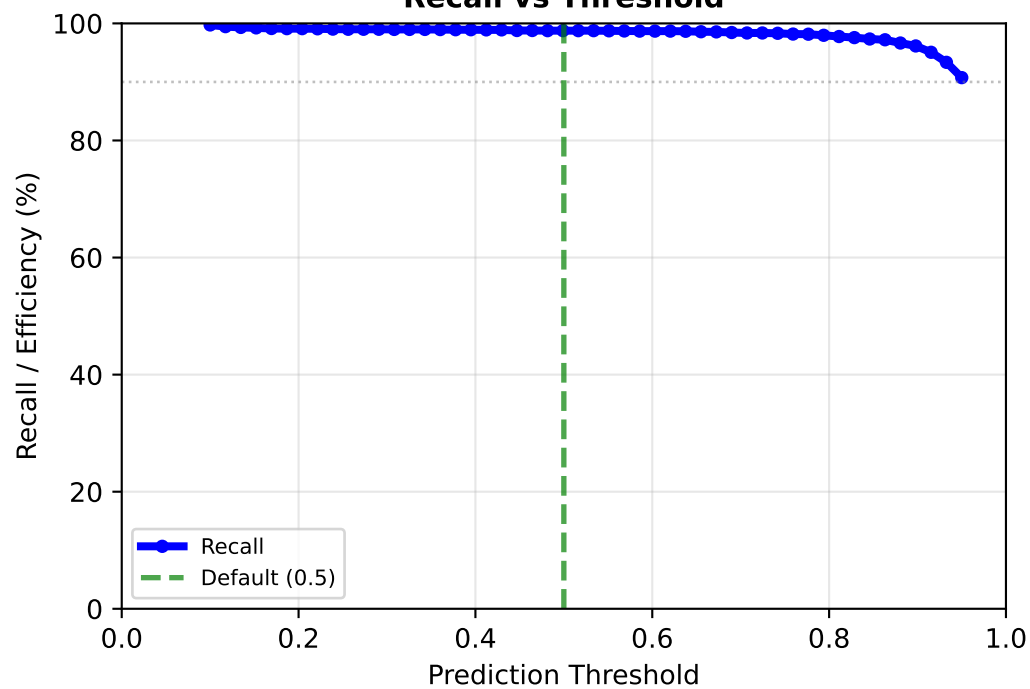


MT Threshold Scan Analysis

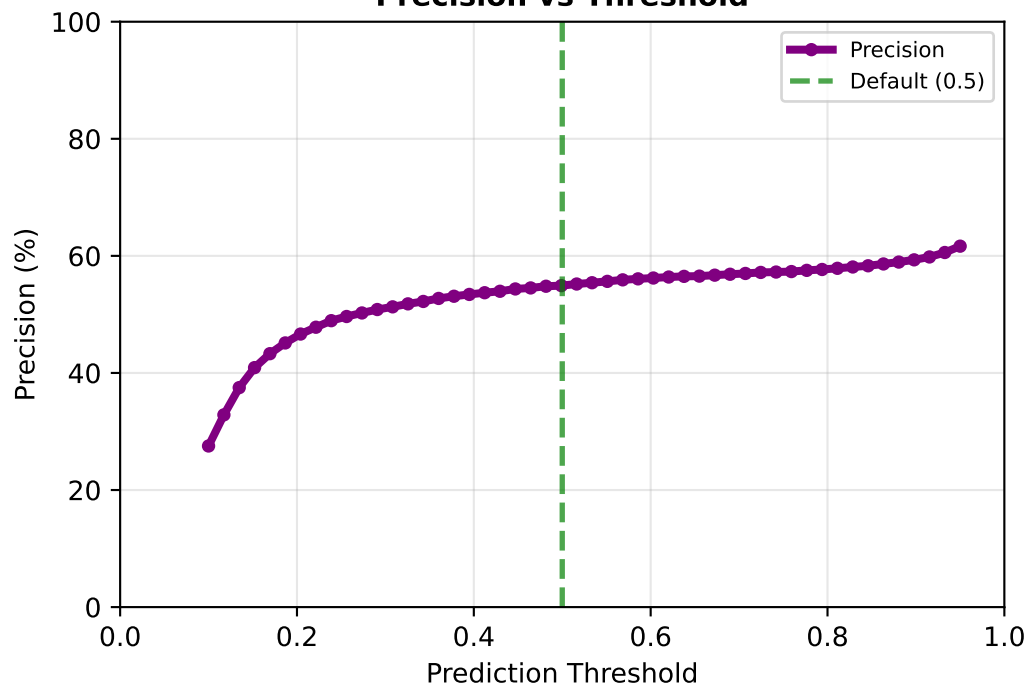
Contamination vs Threshold



Recall vs Threshold



Precision vs Threshold



F1 Score vs Threshold

